



ManageHub

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Outline



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 - **Objectives**
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Introduction

The Problem:

- Small shops still run on notebooks, memory, and guesswork
- Stock, sales, and staff records are scattered, not in one place

Our Solution: One simple system for a shop's daily work:
Existing software is costly and stops working without internet

- **All in one dashboard** - products, billing, staff, and suppliers in one place
- **Works without internet** - a desktop app keeps billing running, then syncs later
- **Talks in plain language** - an AI assistant answers questions and helps with billing
- **Safe for every shop** - each shop's data stays private and separate

Impact: Any shop owner can run their business from a browser - no installation, no IT staff needed

Problem Statement

Critical Gaps in Current Shop Management:

1. Everything Scattered, Nothing Connected

- Stock, sales, customers and suppliers live in separate notebooks or spreadsheets
- No single view of what is really happening in the shop

2. Manual Work, Frequent Mistakes

- Stock runs out with no warning until the shelf is empty
- Cash totals and handovers are slow and easy to get wrong

Problem Statement

Critical Gaps in Current Shop Management:

3. Existing Software Doesn't Fit

- Ready-made ERP or POS tools are too costly or too complex to set up
- Not built for Urdu, local pricing, or how small shops really work

3. No Internet, No Business

- Most systems assume the internet never drops
- A lost connection at the till can stop sales completely
- Missing: **a plain-language assistant that answers questions and helps with billing**

Objectives



Primary Goal: Build one web-based, cloud-hosted system that brings inventory, sales, customers, suppliers, staff, and finance together for small shops - with an offline mode and a plain-language AI assistant

Specific Objectives:

1. Multi-Tenant Web Platform

- Let many shops run on one platform, each with fully private data
- Open and use in any regular browser - nothing to install

2. Complete Retail Operations

- Track inventory, low-stock alerts, and purchase orders
- Run point-of-sale billing that updates stock automatically

Objectives



3. Role-Based Access Control

- Give owners, managers, and cashiers access to only what they need
- Keep sensitive actions restricted to the right roles

3. Offline-First Desktop App

- Keep billing and stock updates working even without internet
- Auto-sync everything back to the cloud once reconnected

3. Agentic AI Assistant

- Answer questions and help with billing in plain English or Urdu
- Always ask a human to confirm actions that touch money or stock

Literature Review

Research Landscape

Manual & Paper-Based Records

- A notebook, calculator, or spreadsheet - free, but does not scale
- No automatic stock tracking or low-stock alerts
- Reports and totals have to be built by hand

Off-the-Shelf ERP & POS Systems

- Odoo, ERPNext, Square, QuickBooks - built for large, global markets
- Powerful, but need a developer to set up and localise
- Not built for Urdu, cash-first sales, or Pakistan's pricing needs

Literature Review

Cloud & SaaS Delivery Models

- A monthly subscription means no server for the shop to maintain
- Shared infrastructure keeps the cost low for each shop
- Server-rendered pages stay simple and consistent on old hardware

AI in Retail & Offline-First Design

- Machine learning is often used to forecast demand and suggest reorders
- Agentic assistants pair a large language model with specialised tools
- Offline-first apps write locally, then sync a queue once back online

Literature Review

Common Design Patterns

- Keyword matching first, then a language model for unclear requests
- A shared router hands each request to a specialised agent

Research gaps + Our Contributions

Gap Area	Current State	Limitations
Offline Support	Rare in affordable systems	Most fail the moment internet drops
Local Language & Pricing	Rarely a first-class feature	Not built for Urdu or local tax norms
Conversational AI	Found mainly in big enterprise tools	Too costly for a small shop's budget
AI on Money/Stock	No human-confirmation step	Risky, fully-automated decisions

Literature Review

Our Contributions

- ✓ **Multi-Tenant SaaS:** one platform, many shops, fully separate data
- ✓ **Truly Offline-First:** desktop app with local storage and auto-sync
- ✓ **Role-Based Access:** owners, managers, and cashiers see only their tools
- ✓ **Human-in-the-Loop AI:** 7 agents, but money and stock always need approval
- ✓ **Web-First, Zero Install:** runs in any browser, updates instantly for every shop
- ✓ **Built for Local Shops:** Urdu support and Pakistan-specific pricing

Methodology

Development Approach

- Built module by module: products and billing first, then HR, offline sync, and AI
- Each module tested before moving to the next one
- Code organised into routes, controllers, views, and helpers

Technology Stack

- Backend: Node.js, Express, and MySQL (SQLite for offline mode)
- Frontend: server-rendered EJS templates and Chart.js dashboards
- AI service: FastAPI, LangChain, and Groq's Llama 3.3 model
- Desktop: Electron, plus Stripe for subscription billing

Methodology



Database Design

- Multi-tenant model: every table carries a shop_id, so data never crosses shops
 - Keys are UUIDs, not auto-increments
 - Lets records made offline merge in later without ID clashes
- Four functional domains, shown in Figure 1
- Confidence in data integrity: one shop's mistake can't affect another

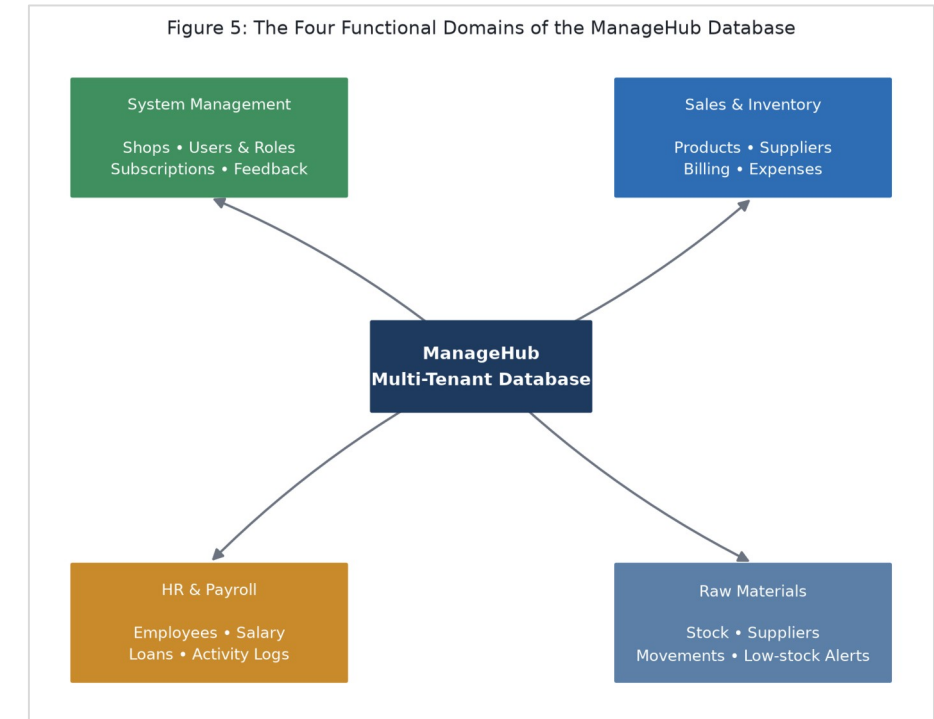


Figure 1: The Four Functional Domains of the ManageHub Database

The Four Domains

Methodology

Customer-Facing Modules

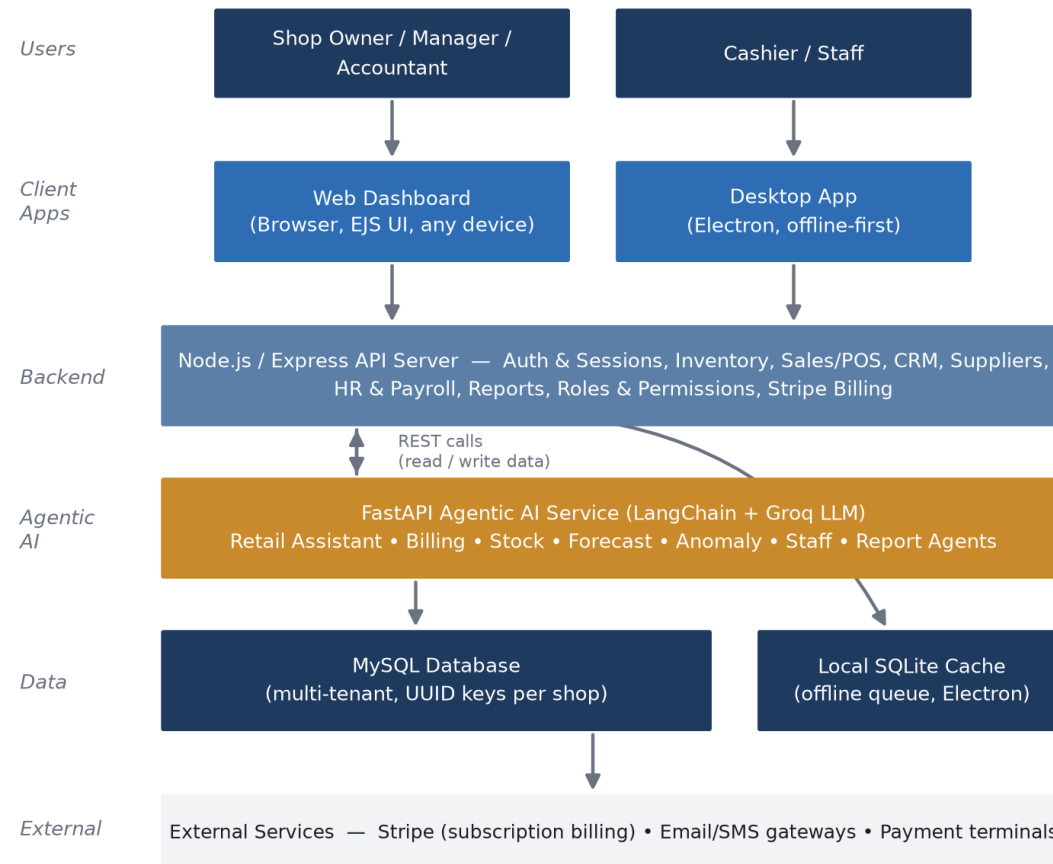
- Inventory & Products - SKUs, stock levels, low-stock alerts
- Point-of-Sale Billing - scan, discount, auto-updates stock
- Customers (CRM) - purchase history and credit limits
- Suppliers - purchase orders and running balances

Back-Office Modules

- Employees & Payroll - salaries, loans, and activity logs
- Reports & Dashboards - charts, PDF and Excel export
- Roles, Feedback & Subscription Billing via Stripe

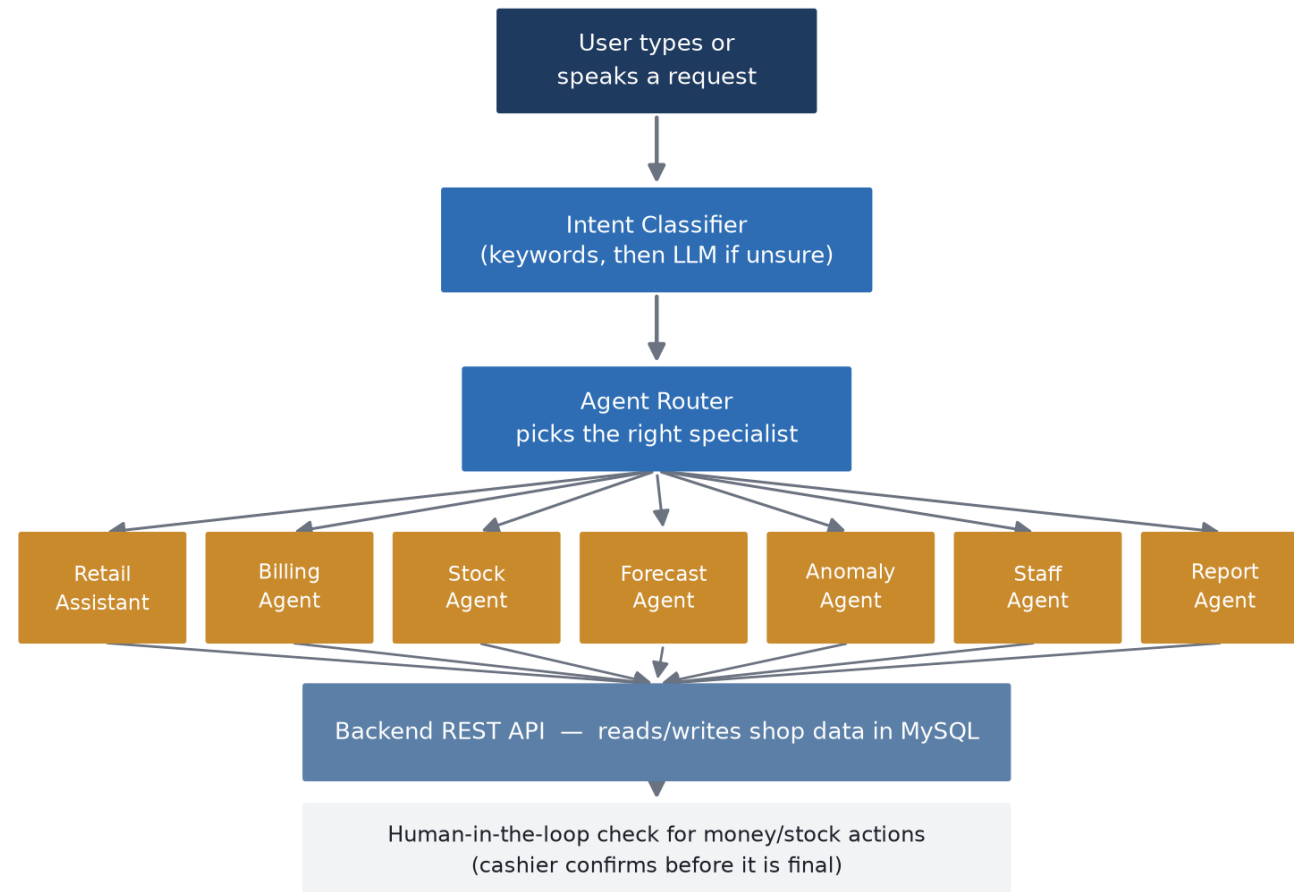
System Architecture

Figure 1: Layered System Architecture of ManageHub



Agentic AI Request Flow

Figure 3: How the Agentic AI Assistant Handles a Request



Results

Data Quality Filtering

Raw detections: 61

Recent events (30 min): 19

Consensus-validated events: 8

False positives filtered: 86.9%

Detection Accuracy

Average initial confidence: 87.4%

Post-consensus confidence: 100%

Event distribution:

- Potholes — 62.5%
- Speed bumps — 37.5%

Aggressive filtering ensures only high-confidence events persist.

All validated events confirmed by ≥ 2 independent devices.

Results

Real-Time System Performance

GPS updates: 100 ms

Backend response: < 1 s

Auto-refresh cycle: 30 s

Platforms tested: Android

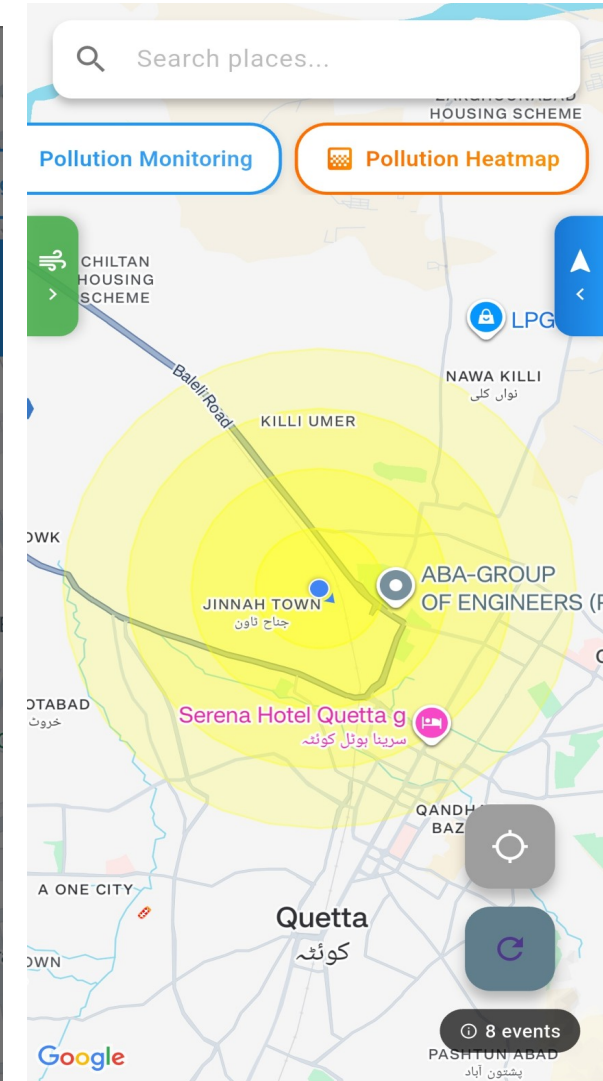
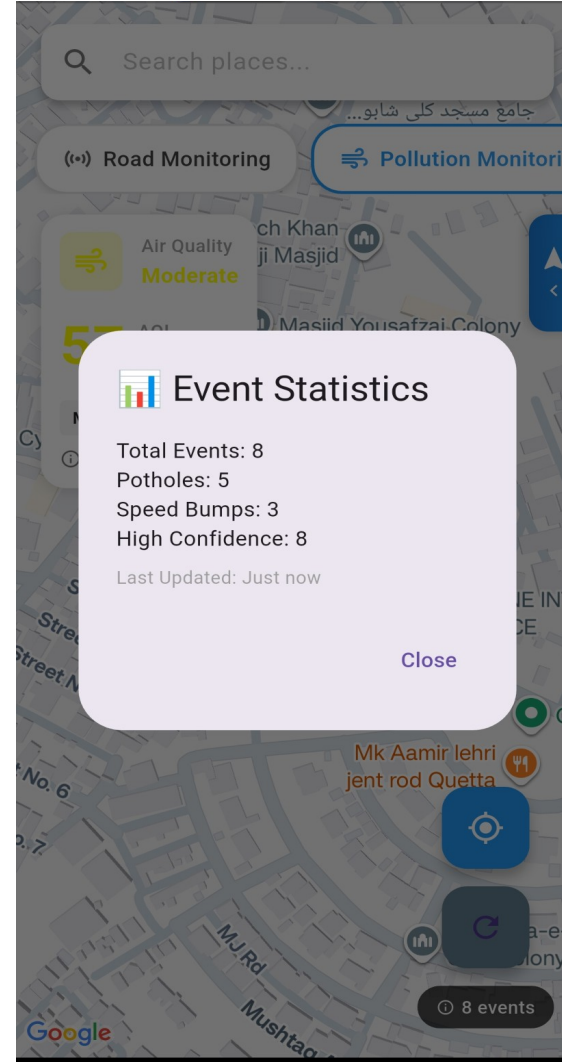
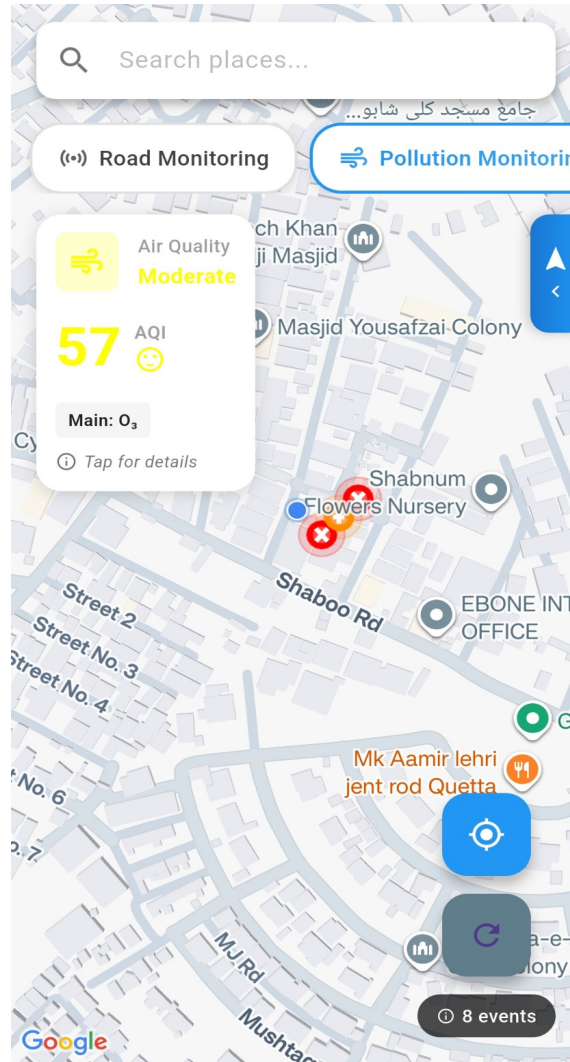
Real-world deployment: Quetta, Pakistan

Key Outcome

Reliable, real-time, crowd-validated road hazard detection using mobile sensors

Results

System in Operation



Project Workload

Abdul Ahad — Core System & Intelligence

- System architecture & service-layer design
- Sensor data processing (accelerometer, gyroscope)
- Road anomaly detection logic
- Consensus validation & noise filtering
- GPS tracking and real-time location handling
- Backend integration (event upload & sync)
- Performance optimization & result analysis

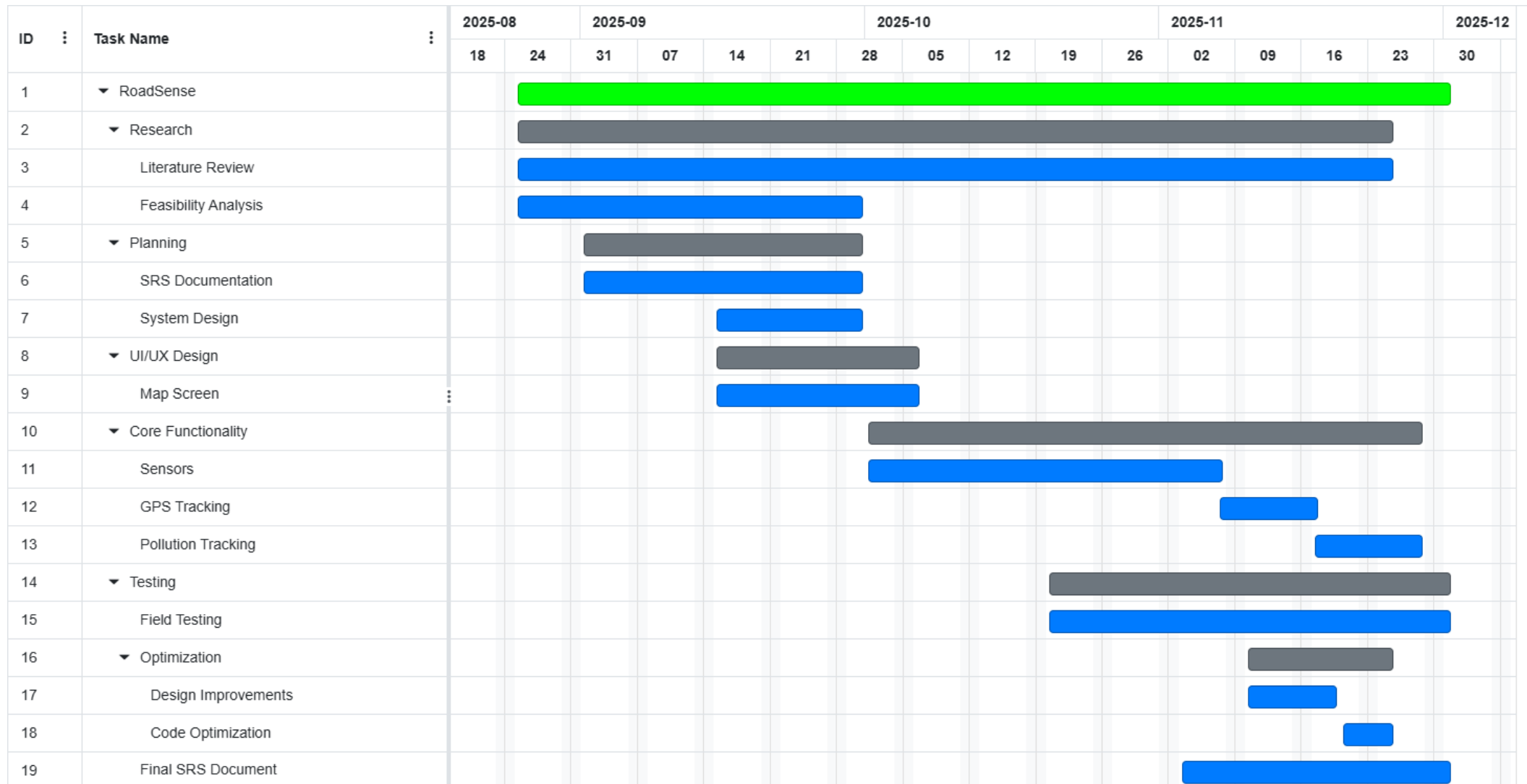
Focus: core logic, data reliability, system performance

Umer Bin Ibrar — UI, Maps & User Interaction

- Flutter UI development & screen flows
- Google Maps integration & visualization
- Hazard markers & pollution overlays
- Navigation interface & user alerts
- Cross-platform consistency (Android & iOS)
- UI/UX improvements and usability testing

Focus: user experience, visualization, interaction design

Gantt Chart



References

- [1] M. Zareei et al., "Machine Learning Model for Road Anomaly Detection Using Smartphone Accelerometer Data," *IEEE Access*, 2025. (98% accuracy, CNN-based, adaptive thresholding)
- [2] P. Mohan et al., "TrafficSense: Rich monitoring of road and traffic conditions using mobile smartphones," Microsoft Research, 2008. (Foundational work, virtual reorientation)
- [3] F. Seraj et al., "RoADS: A road pavement monitoring system for anomaly detection," *Big Data Analytics*, 2014. (Envelope demodulation, 90% accuracy)
- [4] A. Allouch et al., "ManageHub: Smartphone application to estimate road conditions," *IEEE Sensors J.*, 2017. (98.6% accuracy, C4.5 Decision Tree)
- [5] O. Mazari Abdessameud & W. Cherifi, "Federated road surface anomaly detection," *DATA Conf.*, 2024. (LSTM + federated learning)

Thank You!